

National Coverage Determination (NCD)

Positron Emission Tomography (PET) Scans

220.6

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Tracking Information

Publication Number

100-3

Manual Section Number

220.6

Manual Section Title

Positron Emission Tomography (PET) Scans

Version Number**Effective Date of this Version**

03/07/2013

Ending Effective Date of this Version

01/01/2022

Implementation Date

09/03/2013

Description Information

Benefit Category

Diagnostic Tests (other)

Please Note: This may not be an exhaustive list of all applicable Medicare benefit categories for this item or service.

Item/Service Description

Positron Emission Tomography (PET) is a minimally invasive diagnostic imaging procedure used to evaluate metabolism in normal tissues as well as in diseased tissues in conditions

such as cancer, ischemic heart disease, and some neurologic disorders. A radiopharmaceutical is injected into the patient that gives off sub-atomic particles, known as positrons, as it decays. PET uses a positron camera (tomograph) to measure the decay of the radiopharmaceutical. The rate of decay provides biochemical information to on the metabolism of the tissue being studied.

NOTE: This manual section, 220.6 lists all Medicare–covered uses of PET scans. Except as set forth below in cancer indications listed as “Coverage with Evidence Development,” a particular use of PET scans is not covered unless this manual specifically provides that such use is covered. Although this section, 220.6 lists some non-covered uses of PET scans, it does not constitute an exhaustive list of all non-covered uses.

Effective for dates of service on or after March 7, 2013, local Medicare Administrative Contractors (MACs) may determine coverage within their respective jurisdictions for positron emission tomography (PET) using radiopharmaceuticals for their Food and Drug Administration (FDA) approved labeled indications for oncologic imaging.

We emphasize each of the following points:

1. Changing the ‘restrictive’ language of prior PET decisions will not by itself suffice to expand Medicare coverage to new PET radiopharmaceuticals.
2. The scope of this change extends only to FDA-approved indications for oncologic uses of PET tracers.
3. This change does not include screening uses of PET scanning.

The Centers for Medicare & Medicaid Services (CMS) acknowledges the advances relating to the assessment of diagnostic performance and patient safety, as pioneered by the FDA in its regulatory policies and guidelines for diagnostic PET imaging agents and systems during the past decade. We note for completeness that local coverage cannot be in conflict with NCDs or other national policies. Finally, we note that future CMS NCDs, if any, regarding diagnostic PET imaging would not be precluded by this NCD.

Indications and Limitations of Coverage

Cross Reference

See NCD for [SPECT](#) (§220.12).

Claims Processing Instructions

[TN 2750 \(Medicare Claims Processing\)](#)

[TN 11272 \(Medicare Claims Processing\)](#)

[TN 11426 \(Medicare Claims Processing\)](#)

Transmittal Information

Transmittal Number

156

Coverage Transmittal Link

<https://www.cms.gov/Regulations-and-Guidance/Guidance/Transmittals/downloads/R156NCD.pdf> 

Revision History

08/2013 - Effective for dates of service on or after March 7, 2013, local Medicare Administrative Contractors (MACs) may determine coverage within their respective jurisdictions for positron emission tomography (PET) using radiopharmaceuticals for their Food and Drug Administration (FDA) approved labeled indications for oncologic imaging. This revision to the Medicare National Coverage Determinations Manual is a national coverage determination (NCD). NCDs are binding on all carriers, fiscal intermediaries, contractors with the Federal government that review and/or adjudicate claims, determinations, and/or decisions, quality improvement organizations, qualified independent contractors, the Medicare appeals council, and administrative law judges (ALJs) (see 42 CFR section 405.1060(a)(4) (2005)). An NCD that expands coverage is also binding on a Medicare advantage organization. In addition, an ALJ may not review an NCD. (See section 1869(f)(1)(A)(i) of the Social Security Act.) Effective date 03/07/2013. Implementation date 09/03/2013. ([TN 156](#) ) (CR8381)

10/2009 - This change request rescinds and replaces Transmittal 106, dated September 18, 2009. The effective date has been changed to April 3, 2009, and the implementation date has been changed to October 30, 2009. Business Requirements (BR) 6632.6.1 and 6632.6.2 have been revised to clarify that they are subsets of BR 6632.6 and are specific to CED. All other information remains the same. ([TN 108](#) ) (CR6632)

Transmittal 108, dated October 16, 2009, is being rescinded and replaced with Transmittal 120. This transmittal restores language that was inadvertently removed from section 220.6. This transmittal also updates section 220.6.17, per Transmittal 110, Change Request 6753, dated December 18, 2009. All other information remains the same. ([TN 120](#) ) (CR6632)

09/2009 - CMS is adopting a coverage framework that replaces the four-part diagnosis, staging, restaging and monitoring response to treatment categories with a two-part framework that differentiates FDG PET imaging used to inform the initial antitumor treatment strategy from other uses related to guiding subsequent antitumor treatment strategies after the completion of initial treatment. CMS is making this change for all NCDs that address coverage of FDG PET for all oncologic conditions. Effective date 04/06/2009 Implementation date 10/19/2009. ([TN 106](#) ) (CR6632)

04/2005 - Covered as an adjunct test for detection of pre-treatment metastases (i.e., staging) in newly diagnosed cervical cancer subsequent to conventional imaging that is negative for extra-pelvic metastasis. For all remaining indications that are subject of this NCD, as well as other indications for cervical cancer not specifically mentioned, covered only when providers are participating in, and patients are enrolled in, an approved FDG PET clinical study, or an FDG PET clinical trial meeting FDA category B IDE exemption status. Effective date 01/28/2005. Implementation date 4/18/2005. ([TN 31](#)) (CR 3741)

06/2003 - Expanded coverage for noninvasive imaging of perfusion of heart using FDA-approved Ammonia N-13 tracer, and restaging of recurrent or residual thyroid cancers of follicular cell origin that have been previously treated by thyroidectomy and radioiodine ablation and have a serum thyroglobulin >10ng/ml and negative I-131 whole body scan. Also maintained national noncoverage determinations for all uses of FDG-PET for soft tissue sarcoma and dementia and neurogenerative diseases. Effective and implementation dates 10/01/2003. ([TN 171](#)) (CR 2687)

05/2002 - Provided coverage for breast cancer and broader coverage for myocardial viability. Effective and implementation dates 10/01/2002. ([TN 156](#)) (CR 2138)

11/2001 - Clarified types of allowable PET scanners. Effective and implementation dates 01/01/2002. ([TN 147](#)) (CR 1886) Reduction in coverage announced in Federal Register, 66 FR 58741.

04/2001 - Expanded coverage to include diagnosis and staging/restaging of non-small cell lung cancer, esophageal cancer, colorectal cancer, lymphoma, melanoma, head and neck cancers, myocardial viability, and refractory seizures. Effective and implementation dates July 1, 2001. ([TN 136](#)) (CR 1603)

04/1999 - Added coverage for evaluation of recurrent colorectal cancer in patients with rising levels of carcinoembryonic antigen, staging of lymphoma (both Hodgkins and non-Hodgkins) when PET scan substitutes for gallium scan, and detection of recurrent melanoma. Effective July 1, 1999. (TN 113) (CR 861)

02/1999 - Added coverage for characterization of solitary pulmonary nodules and initial staging of lung cancer. Effective date 1/1/1998. (TN 106)

05/1995 - Provided limited coverage for use in noninvasive imaging of perfusion of heart for diagnosis and management of patients with known or suspected coronary artery disease. Effective date 03/14/1995. (TN 76)

National Coverage Analyses (NCAs)

This NCD has been or is currently being reviewed under the National Coverage Determination process. The following are existing associations with NCAs, from the National Coverage Analyses database.

- [Original Consideration for Positron Emission Tomography \(FDG\) \(CAG-00065N\)](#) 
- [First reconsideration for Positron Emission Tomography \(NaF-18\) to Identify Bone Metastasis of Cancer \(CAG-00065R\)](#) 
- [Second reconsideration for Positron Emission Tomography \(NaF-18\) to Identify Bone Metastasis of Cancer \(CAG-00065R2\)](#) 
- [Second reconsideration for Positron Emission Tomography \(CAG-00065R2\)](#) 
- [Original Consideration for Positron Emission Tomography \(FDG\) for Alzheimer's Disease/Dementia \(CAG-00088N\)](#) 
- [First reconsideration for Positron Emission Tomography \(FDG\) and Other Neuroimaging Devices for Suspected Dementia \(CAG-00088R\)](#) 
- [Original Consideration for Positron Emission Tomography \(FDG\) for Breast Cancer \(CAG-00094N\)](#) 
- [Original Consideration for Positron Emission Tomography \(FDG\) for Brain, Cervical, Ovarian, Pancreatic, Small Cell Lung, and Testicular Cancers \(CAG-00181N\)](#) 
- [First reconsideration for Positron Emission Tomography \(FDG\) for Solid Tumors \(CAG-00181R\)](#) 
- [Second reconsideration for Positron Emission Tomography \(FDG\) for Cervical Cancer \(CAG-00181R2\)](#) 
- [Third reconsideration for Positron Emission Tomography for Initial Treatment Strategy in Solid Tumors and Myeloma \(CAG-00181R3\)](#) 
- [Fourth reconsideration for Positron Emission Tomography \(FDG\) for Solid Tumors \(CAG-00181R4\)](#) 
- [Original Consideration for Positron Emission Tomography \(FDG\) for Infection and Inflammation \(CAG-00382N\)](#) 
- [Original Consideration for Beta Amyloid Positron Emission Tomography in Dementia and Neurodegenerative Disease \(CAG-00431N\)](#) 

Additional Information

Other Versions

Title	Version	Effective Between	
Positron Emission Tomography (PET) Scans - RETIRED	7	04/10/2023 - N/A	View
Positron Emission Tomography (PET) Scans	6	01/01/2022 - 04/10/2023	View
Positron Emission Tomography (PET) Scans	5	03/07/2013 - 01/01/2022	You are here
Positron Emission Tomography (PET) Scans	4	04/03/2009 - 03/07/2013	View
PET Scans	3	01/28/2005 - 04/03/2009	View
PET Scans	2	10/01/2003 - 01/28/2005	View
PET Scans	1	10/01/2002 - 10/01/2003	View